

Physics-Informed Digital Twin Integrating Raman Distributed Fibre Optic Sensing for Predictive Maintenance of Utility-Scale Solar Farm Inverters

Reliable operation of utility-scale photovoltaic (PV) systems is critical to the global renewable energy transition, with solar PV accounting for 42% of the world's electricity generation today, projected to reach 50% by 2030¹. Within plants, power electronic inverters remain among the most failure-prone components responsible for 43% of reported system failures and 36% of downtime-related energy losses². String inverters, deployed widely across utility-scale farms as a distributed architecture with one unit per panel row, are particularly exposed to thermal-electrical degradation mechanisms such as insulated gate bipolar transistor electrical ageing, thermal-interface degradation, and internal thermal path deterioration, driving substantial O&M costs and yield loss that scales with plant size.

Existing condition monitoring techniques remain largely reactive, relying on point sensors that detect degradation only after it has progressed, which cannot localise where within or across inverter units it originates^{3,4}. This research proposes a physics-informed digital twin coupling a reduced-order string inverter thermal model with simulated Raman distributed temperature sensing (DTS), enabling spatial comparison between predicted and measured thermal behaviour across multiple inverter units. Machine learning applied to these points will be used to classify degradation states and generate predictive maintenance alerts ahead of failures, enabling earlier, spatially localised fault detection.

Keywords: Digital twin, photovoltaic systems, string inverter, Raman distributed temperature sensing, predictive maintenance, condition monitoring, machine learning.

References

1. International Energy Agency (IEA), "Global electricity demand is set to grow strongly to 2030, underscoring need for investments in grids and flexibility," *IEA News*, 2026.
2. T. J. Formica, H. A. Khan, and M. G. Pecht, "The Effect of Inverter Failures on the Return on Investment of Solar Photovoltaic Systems," *IEEE Access*, vol. 5, pp. 21336–21343, Sep. 2017, doi: 10.1109/ACCESS.2017.2753246.
3. X. Bao and L. Chen, "Recent Progress in Distributed Fiber Optic Sensors," *Sensors*, vol. 12, no. 7, pp. 8601–8639, Jul. 2012, doi: 10.3390/s120708601.
4. B. Guzowski, M. Lakowski, and D. Bobinski, "Hotspot Detection in Photovoltaic Modules with Fiber Bragg Grating and Brillouin Distributed Temperature Sensors," *Energies*, vol. 18, no. 23, Art. 6117, 2025, doi: 10.3390/en18236117.